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PHYSICS Unit-II

Polarisation Chapter 3 Unit-II

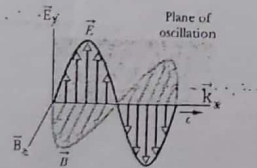
- Introduction, plane of polarization & plane of vibration
- Production of plane polarized light by different methods
- Brewster's law, Malus law (definition only)
- Optic axis, Principal section
- Double refraction
- Double refracting uniaxial crystals
(positive & Negative uniaxial crystals and their wave fronts)
- Nicol prism (Construction, Action and Limitation)
- Quarter and half wave plates,
- Production of plane, circularly and elliptically polarized lights
- Specific rotation, Laurent's half shade polarimeter.

Dr. Pragati sharma Applied Physics-I

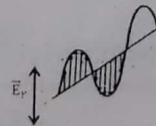
I-Sem
Dr. Pragati

According to electromagnetic theory of light

A light wave consists of electric and magnetic field vectors $\vec{E}$ & $\vec{B}$ vibrating in mutually perpendicular planes and also $\perp$ to the direction of propagation of light ($\vec{k}$).

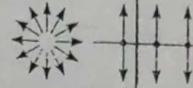


By convention, the electric vector represents the light vector.



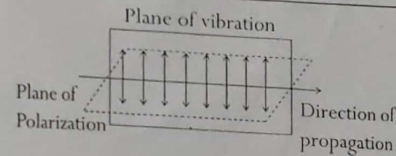
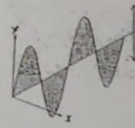
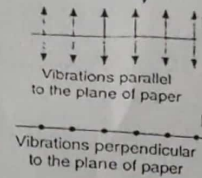
Un-polarized light

has large number of vibrations of light vector in all possible planes, with equal probability, at right angles to the direction of propagation.



Plane polarized light

has only one plane of vibrating vector along a fixed line, at right angles to the direction of propagation.



Plane of vibration

is a plane that contains the direction of vibration.

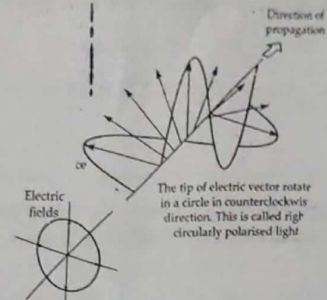
Plane of Polarization

is a plane perpendicular to the plane of vibration & is passing through the direction of propagation.

Circularly or Elliptically polarized light is resultant of two plane polarized waves differing in phase by $\pi/2$.

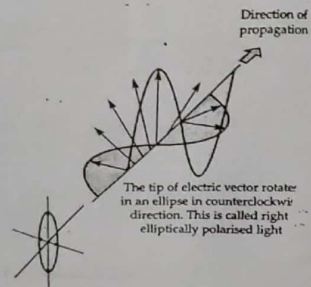
Circularly polarized light

If two plane polarized waves have equal amplitudes & differing in phase by $\pi/2$, the result is a circularly polarized light. The resultant electric vector appears to be rotating circularly.



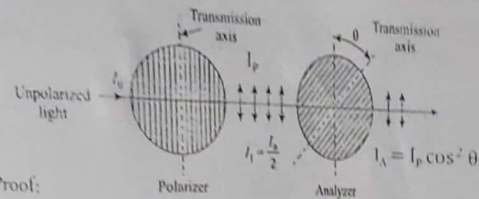
Elliptically polarized light

If two plane polarized waves have unequal amplitudes & differing in phase by $\pi/2$, the result is an elliptically polarized light. The resultant electric vector appears to be rotating elliptically.



Malus law gives the Intensity of Polarized light

"When polarized light is passed through analyzer its intensity, I_A varies as square of the cosine of angle between plane of transmission of analyzer & plane of polarizer" $I_A \propto \cos^2 \theta$



Proof:

If 'a' represents the amplitude of incident Plane Polarized Light and θ the angle b/n plane of vibration of PPL & optic axis of analyzer, then $a \sin \theta$, the component perpendicular to it plane of analyzer,

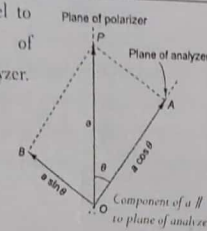
while $a \cos \theta$ is the component parallel to plane of analyzer. The Intensity of transmitted component through the analyzer.

$$I_A = a^2 \cos^2 \theta$$

Where $a^2 = I_p$ is intensity of PPL

$$I_A = I_p \cos^2 \theta$$

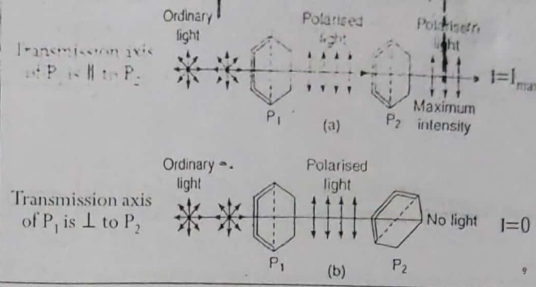
$I_A \propto \cos^2 \theta$, which is a Malus law



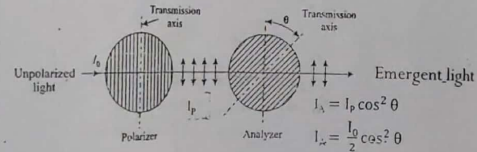
- For $\theta = 0^\circ$ or 180° $I_A = I_p$
When plane of Polarizer & Analyzer are Parallel, light of maximum intensity is transmitted.
- For $\theta = 90^\circ$ $I_A = 0$
When plane of Polarizer & Analyzer are Perpendicular, No light transmitted across the analyzer

Transverse nature of light wave by Polarization

It experimentally demonstrated the phenomena of polarization of light & the fact that light wave is transverse in nature, ordinary light is allowed to pass through a pair of Tourmaline crystal (P_1 & P_2), where P_1 & P_2 act as Polarizer & Analyzer respectively.



Q: Two Nicol Prisms are oriented with their principal planes making an angle of 60° . What percentage of initial un-polarized light will pass through the system?



Sol: $I_A = I_p \cos^2 \theta$

$$I_A = \frac{I_0}{2} \cos^2 60^\circ = \frac{(0.25)I_0}{2} = (0.125) I_0$$

$$12.5\% \text{ of } I_0$$

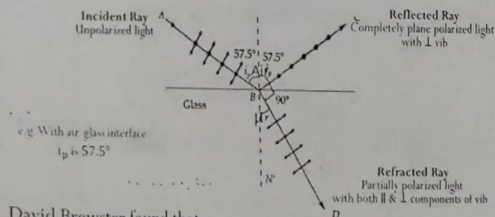
12.5% of initial un-polarized light will pass through the system.

Methods of producing a plane polarized light

- by Reflection,
- by Refraction,
- by Double Refraction (Nicol prism),
- by Selective Absorption, &
- by Scattering

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Polarization by Reflection: Brewster law (1811)



David Brewster found that

When light is incident at certain angle of incidence $\angle i_p$ called polarizing angle, the reflected part of light becomes completely plane polarized with vibrations $\perp$ to the plane, while the refracted part remains partially polarized containing both $\parallel$ & $\perp$ components of vibrations.

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$$\mu = \tan i_p$$

The value of polarizing angle $\angle i_p$ depends upon the refractive index of reflecting medium. This is Brewster law. A direct outcome from the law is that the angle between reflected & refracted beam is 90°

$$\angle i_p + \angle r = 90^\circ$$

Proof: From Snell's law $\mu = \frac{\sin i_p}{\sin r}$

and from Brewster law $\mu = \tan i_p$

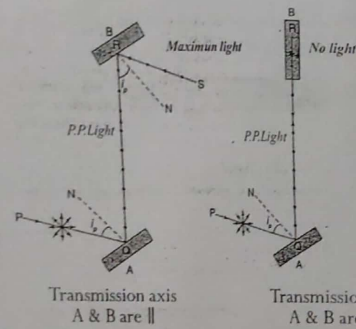
$$\frac{\sin i_p}{\sin r} = \frac{\sin i_p}{\cos i_p}$$

$$\sin r = \cos i_p$$

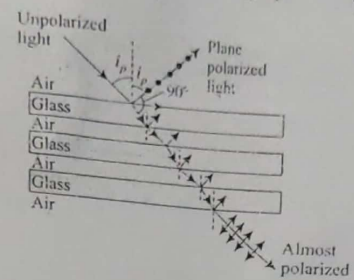
$$\sin r = \sin(90 - i_p)$$

$$\angle r = 90^\circ - \angle i_p \quad \text{and hence} \quad \angle r + \angle i_p = 90^\circ$$

Biot's polariscope: It is an instrument used to produce & detect Plane polarized light using the concept of Polarization by Reflection



Polarization by Refraction:
When light is Refracted through piles of plates (m)



With degree of polarization

$$p = \frac{m}{m + \left(\frac{2\mu}{1-\mu^2}\right)^2}$$

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Q: Light travelling in water ($\mu_w = 1.33$) is incident on a plate of glass ($\mu_g = 1.53$). At what angle of incidence will the light be plane polarized.

Sol: $\mu = \tan i_p$

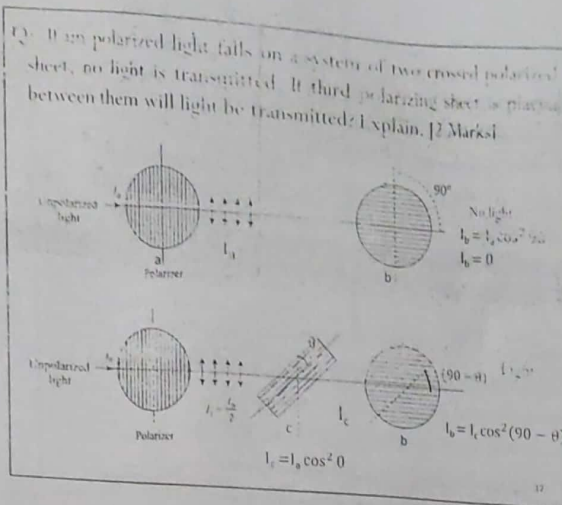
$$\frac{\text{Water } (\mu_w = 1.33)}{\text{Glass } (\mu_g = 1.53)}$$

In the present question $\mu = \frac{\mu_{\text{water}}}{\mu_{\text{glass}}} = \frac{1.33}{1.53} = 1.15$

Hence $i_p = \tan^{-1} 1.15$

$i_p = 49^\circ 12'$

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Solution When third sheet 'c' is inserted between 'a' & 'b' but not parallel to either, then from Malus law, there will be some intensity of light at the output of 'b'

$$I_b = I_c \cos^2 (90^\circ - \theta)$$

$$I_b = I_c \sin^2 \theta$$

$$I_b = I_a \cos^2 \theta \sin^2 \theta$$

Transmission of light through 'b' depends on the orientation of 'c'

- If $\theta = 0$, two polarizing sheets 'a' & 'b' are in crossed position

$$I_b = I_a \cos^2 90 = 0 \quad \text{No light is transmitted}$$

$$I_b = 0$$

- If $\theta = 45^\circ$

$$I_b = I_a \left(\frac{1}{\sqrt{2}} \right)^2 \left(\frac{1}{\sqrt{2}} \right)^2 = \frac{I_a}{2} \left(\frac{1}{2} \right) = \frac{I_a}{4}$$

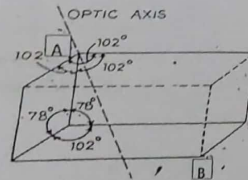
Double refracting crystal (e.g. Calcite, Quartz etc.)

There are certain crystals found in nature where in when a ray of unpolarized light is incident on a crystal at certain angle of incidence $\angle i$, it splits into two refracted components which are plane polarized named as p-ray & e-ray. Each component is propagating in different directions.

Calcite crystal (RHELAND SPAR)

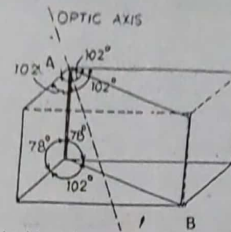
- It is a double refracting, 'birefringent' crystal. Light entering the crystal splits into two polarized components.
- Its crystalline form is calcium carbonate, CaCO_3 .
- It is Colorless & transparent to visible & U.V.
- It occurs in nature in various forms, all of which on cleavage give Rhombohedral in Hexagonal system ($\alpha=\beta=\gamma \neq 90^\circ$).

Geometry of Calcite crystal: Its structure is Rhombohedral ($\alpha=\beta=\gamma \neq 90^\circ$). The Six faces of Rhombohedron are parallelograms each having angles 78° & 102° .



At diametrically opposite corners A & B, three faces meet with obtuse angles (102° each), are known as **Blunt corners**, while at rest of the six corners, two faces meet at acute angle (78°) & one at obtuse angle (102°).

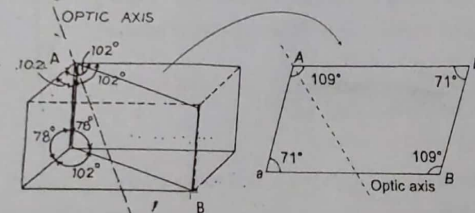
Optic axis is a line passing through one of the blunt corners and inclined equally to all the three edges which meet gives the direction of optic axis. Any line parallel to optic axis is again an optic axis. $\therefore$ optic axis is not a line, it is a direction.



Optic axis of a doubly refracting crystal can also be defined as the axis along which the speed of o-ray & e-ray are equal.

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Principal section is a plane that contains an optic axis and is perpendicular to opposite faces. It cuts the crystal surface in a parallelogram with 71° and 109°.

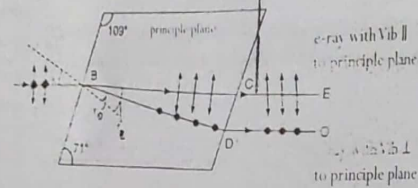


When look straight down on a rhomb shaped calcite crystal, an object beneath it is seen double. On rotating the crystal, one image stays nearly stationary and the other moves around it.

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Double Refraction (Birefringence)

Double refraction is an optical property in certain crystals



When an unpolarized light is incident on a calcite crystal at certain angle of incidence w.r.t. optic axis, it splits into two refracted components which are plane polarized named as ordinary ray (o-ray) & extraordinary ray (e-ray). Each propagating in different directions, making an angle of refraction as $\angle r_o$ and $\angle r_e$ resp.

1. In o-ray the vibrations of electric vector are $\perp$ to principle plane while in e-ray, they are $\parallel$ to principle plane.
2. o-ray obeys the law of refraction, it always lies in plane of incidence while e-ray does not obey the law of refraction.

$$\mu_o = \frac{\sin i}{\sin r_o} \quad \text{constant}$$

$$\mu_e = \frac{\sin i}{\sin r_e} \quad \text{varies}$$

The difference, $(\mu_o - \mu_e)$ is known as birefringence.

The refractive index corresponding to o-ray is constant for all angles of incidence (same in all directions), while for e-ray it varies with angle of incidence (direction dependent).

3. since $v_o = \frac{c}{\mu_o}$ v_o remains Constant

and $v_e = \frac{c}{\mu_e}$ v_e Varies with $\angle i$ w.r.t. optic axis

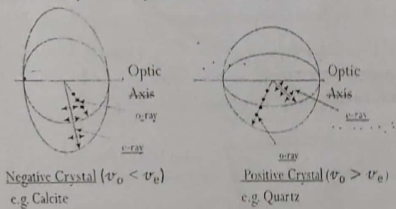
The velocity of o-ray remains same in all directions while that of e-ray varies with directions of incident light w.r.t the optic axis.

In negative crystals e.g. calcite, the velocity v_e is greatest in a direction $\perp$ to optic axis, while in positive crystals like Quartz the velocity v_e is least in a direction $\perp$ to optic axis. However, along optic axis only, $v_o = v_e$. Thus optic axis can also be defined as the axis along which the speed of o-ray & e-ray are equal.

Once o-ray & e-ray emerge from the crystal they have same velocity, differ only in their direction of propagation & plane of polarization.

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4. From Huygen's theory of double refraction, wave front corresponding to o-ray spread out as Sphere as v_o is constant, while for e-ray it is Ellipsoidal because v_e Varies.



Along the optic axis the two rays travel with the same velocity ($v_o = v_e$), the sphere & ellipsoid touch each other at a point which lies on the optic axis of the crystal.

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Types of doubly refracting crystals

Uniaxial crystals: Crystals in which there is only one optic axis i.e. only one direction along which the two refracted rays travel with the same velocity e.g. Calcite, Tourmaline, and Quartz. Such crystals are of two varieties.

Negative uniaxial e.g. Calcite

$$v_o < v_e$$

v_e is greatest in a direction $\perp$ to optic axis. Ellipsoid lies outside Sphere. (refer fig)

Positive uniaxial e.g. Quartz

$$v_o > v_e$$

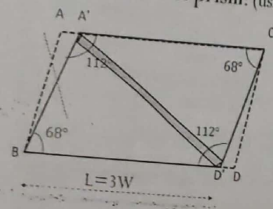
v_e is least in a direction $\perp$ to optic axis. Ellipsoid lies inside Sphere. (refer fig)

In Biaxial crystals there are two optic axis or direction along which the velocities are the same e.g. Topaz, aragonite, etc.

Nicol prism

is a device consists of doubly refracting crystal, used for Producing & Analyzing plane polarized light. It eliminates one of the two polarized light by Total Internal Reflection (TIR) at the surface of thin film of Canada balsam.

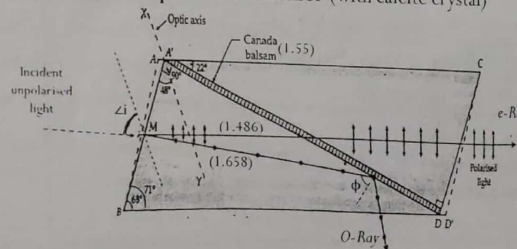
Construction of Nicol prism: (using Calcite crystal)



Ground 71° to 68° & 109° to 112°
Cut,
Cemented by Canada balsam
 $[\mu_o > \mu_{CB} > \mu_e]$
Enclosed

- Take a calcite crystal with sides ABCD about three times long as its width, $l \approx 3W$
- Ground its end faces AB & CD such that angles in the principal section changes from 71° & 109° to 68° & 112° respectively.
- Cut apart the crystal along the plane A'D'.
- The two cut surfaces are ground and polished optically flat and are then cemented together by Canada balsam which is a transparent liquid for sodium light. The refractive index of Canada balsam (1.55) lies between the refractive index of calcite for o-ray (1.658) & e-ray (1.486)
- The whole assembly is then enclosed in a tube blackened-inside.

Action of a Nicol prism as Polarizer (with calcite crystal)



When incident unpolarised light hits the face A'B, at certain angle $\angle i$ w.r.t. the optic axis, it splits into two refracted components, o-ray & e-ray which are plane polarized. The vibrations of electric vector in o-ray are $\perp$ to principle plane while in e-ray they are $\parallel$ to principle plane.

$[\mu_o > \mu_{CB} > \mu_e]$ The refractive index of Canada balsam (1.55) is less than the refractive index of o-ray (1.658) & more than that of e-ray (1.486) for a calcite crystal.

When o-ray reaches the boundary of Canada balsam, it passes from optically denser to rarer medium. Since the length of the crystal is large, the angle of incidence of o-ray at Calcite-Canada balsam surface becomes greater than the critical angle ($\phi > \phi_c$). Under these conditions the o-ray gets totally reflected at Calcite-Canada balsam interface and finally absorbed by the tube containing the crystal. Whereas e-ray passes the interface of Cal-CB from rarer to denser medium & is transmitted. Thus emergent light from Nicol prism is plane polarized e-ray with vibrations $\parallel$ to principal section.

Quest: Show that if angle of incidence for o-ray at calcite-Canada balsam interface ϕ is greater than 69° , then o-ray gets TIR.

Ans: From Snell's law at critical angle:

$$\mu_o \sin \phi_c = \mu_{CB} \sin 90$$

$$\phi_c = \sin^{-1} \frac{\mu_{CB}}{\mu_o}$$

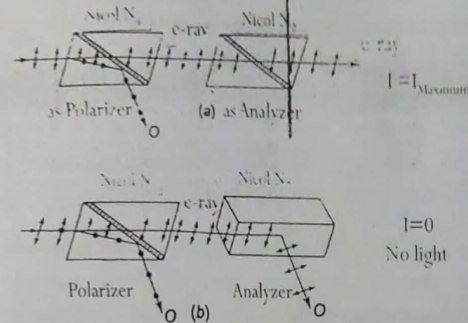
$$\therefore \phi_c = \sin^{-1} \frac{1.55}{1.65}$$

Thus the critical angle for o-ray at Calcite-Canada balsam interface is

$$\phi_c = 69^\circ$$

For TIR, the angle of incidence should be greater than critical angle $\phi > \phi_c$. Hence if the angle of incidence for o-ray at calcite-Canada balsam interface is greater than 69° , the o-ray gets TIR.

Polarization by Double Refraction: Justification



Ray emerging from N_2 prism is PPL e-ray with vibrations $\parallel$ to principal plane.

Nicols in parallel position ($N_2 \parallel N_1$)

When Principal sections of two Nicols ' N_1 ' & ' N_2 ' are placed parallel to each other, the e-ray incident on N_2 comes out of it again as e-ray only $\Rightarrow$ The light emerges out with maximum intensity

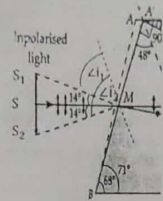
Nicols in crossed position ($N_2 \perp N_1$)

When analyzer is rotated such that the principal section of N_2 becomes $\perp$ to that of N_1 . The e-ray before entering N_2 is plane polarized with vibs $\parallel$ to the principle section. However as this ray enters N_2 , its vibrations appear $\perp$ w.r.t. to the principal section of N_2 . This makes e-ray to behave as o-ray within N_2 and hence will be lost by TIR at the calcite-balsam surface.

$\Rightarrow$ Light will not emerge out from N_2 .

Limitations with Nicol Prism

Nicol prism can act as a polarizer to get a plane polarized light only when the incident light is slightly divergent or convergent, i.e. beam is limited to incident within the angle of 28° zone lying b/n the extreme rays within (S_1MS_2) .



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If incident ray makes an angle

- smaller than $\angle i_1$, e-ray becomes more and more parallel to the optic axis, its refractive index becomes nearly equal to that of o-ray in Calcite crystal. This causes e-ray also to suffer TIR like o-ray & hence no light will emerge out of Nicol prism.
- greater than $\angle i_2$, o-ray will strike the CB surface at an angle less than critical angle ($\phi < \phi_c$). The o-ray will be not experience TIR at the surface & hence get transmitted. Thus light emerging from Nicol prism will not be plane polarized.

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Retardation plate

It is a device for producing & detecting circularly and elliptically polarized light. It is a plate of certain thickness cut out from a doubly refracting uniaxial crystal with its optic axis $\parallel$ to the refracting face. It is used to produce a definite value of path difference Δ between e-ray and o-ray on emergence. There are two types of retardation plates depending on the path difference they introduce; Quarter wave plate (QWP) and Half wave plate (HWP).

Limitation of Retardation plates: The Retardation plates are wavelength dependent. For sources of different wavelengths the same retardation plate can not be effective equally. They are applicable for one particular wavelength only because to produce a required path difference their thickness is λ dependent.

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Quarter wave plate (QWP)

is a plate of thickness 't' which produces a path difference Δ of $\lambda/4$ or phase difference of $\pi/2$ between e-ray and o-ray on emergence.

$$\Delta = \lambda/4$$

$$(\delta\mu)t = \lambda/4$$

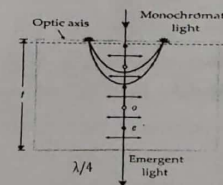
For Negative crystal e.g. Calcite

$$(\mu_o > \mu_e)$$

$$(\mu_o - \mu_e)t = \frac{\lambda}{4}$$

$$t = \frac{\lambda}{4(\mu_o - \mu_e)}$$

$$\text{For Positive crystal e.g. Quartz: } t = \frac{\lambda}{4(\mu_e - \mu_o)}$$



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Half wave plate (HWP)

is a plate of thickness 't' which produces a path difference Δ of $\lambda/2$ or phase difference of π between e-ray & o-ray on emergence.

$$\Delta = \lambda/2$$

$$(\delta\mu) t = \lambda/2$$

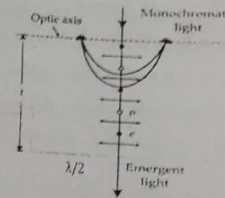
For Negative crystal e.g. Calcite

$$(\mu_o > \mu_e)$$

$$(\mu_o - \mu_e)t = \frac{\lambda}{2}$$

$$t = \frac{\lambda}{2(\mu_o - \mu_e)}$$

For Positive crystal e.g. Quartz $t = \frac{\lambda}{2(\mu_e - \mu_o)}$

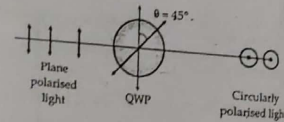


Use of Retardation plates

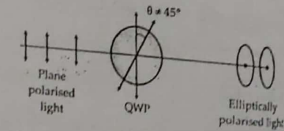
1. Use of QWP

It converts plane polarized light to either circularly or elliptically polarized. If incident P.P. light makes an angle:

$\theta = 45^\circ$ with optic axis, the emergent light is Circularly polarized.



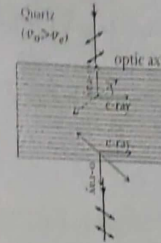
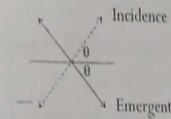
$\theta \neq 45^\circ$ it is Elliptically polarized.



2. Use of HWP:

It rotates plane of polarization of plane polarized light. It is used in Polarimeter for the construction of Laurents Half shade device to improve its sensitivity.

If plane polarized light incidence on HWP making an angle θ with optic axis, the emergent light is also plane polarized but with vibrations inclined at an angle 2θ with respect to its initial vibration.



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HWP rotates the plane of vibration of incident plane polarized light by an angle of 2θ (refer fig from previous slide) Explanation !

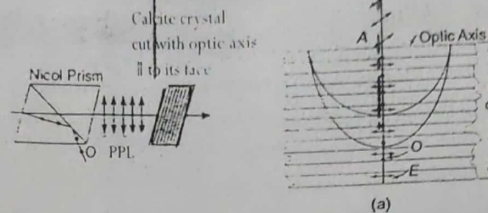
Let a plane polarized light is incident on a HWP of quartz with plane of vibration making an angle θ with optic axis. As it enters the quartz it splits into two components (e-ray & o-ray). The component with vibrations $\parallel$ to optic axis is e-ray, while the component with vibrations $\perp$ to optic axis is o-ray.

Inside Quartz crystal, o-rays travel faster ($v_o > v_e$), hence on emergence through HWP, it gains a phase difference of π over e-ray. This causes the direction of vibration of o-ray to reverse, thus makes the resultant vibration at 2θ with respect to the initial vibration.

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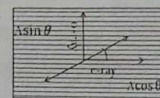
Production of plane, circularly & elliptically pol. lights

Let a PPL from Nicol prism falls normally on a calcite crystal.



Let A be the maximum amplitude of incident PPL which makes an angle θ with the optic axis. As it enters calcite crystal, the plane polarized light splits into two components, o-ray & e-ray.

- e-ray component has vibs $\parallel$ to optic axis with amplitude as $A \cos \theta$, while
- o-ray component has vibs $\perp$ to optic axis with amplitude as $A \sin \theta$



Let $A \cos \theta = a$ and $A \sin \theta = b$

Inside the crystal, these two rays travel with different velocities ($v_e \neq v_o$). Let on emergence from the crystal, a path difference of Δ or a phase difference of φ is introduced b/n them depending on the thickness of the crystal. Thus the two simple harmonic waves with phase difference of φ is given by.

$$x = a \sin(\omega t + \varphi) \quad \dots \text{for e-ray} \dots (1)$$

$$y = b \sin \omega t \quad \dots \text{for o-ray} \dots (2)$$

from

$$\frac{x}{a} = \sin(\omega t + \phi)$$

$$\frac{x}{a} = \sin \omega t \cos \phi + \cos \omega t \sin \phi \quad \text{----- (3)}$$

Where $\sin \omega t$ and $\cos \omega t$ from eq. (2) are

$$\sin \omega t = \frac{y}{b} \text{ and } \cos \omega t = \sqrt{1 - \left(\frac{y^2}{b^2}\right)} \quad \text{----- (4)}$$

Rewriting eq. (3)

$$\frac{x}{a} = \frac{y}{b} \cos \phi + \sqrt{1 - \left(\frac{y^2}{b^2}\right)} \sin \phi$$

$$\frac{x}{a} - \frac{y}{b} \cos \phi = \sqrt{1 - \left(\frac{y^2}{b^2}\right)} \sin \phi \quad \text{----- (5)}$$

Squaring & rearranging above eq. gives a general eq. of an ellipse

$$\left(\frac{x}{a}\right)^2 - \frac{2xy}{ab} \cos \phi + \left(\frac{y}{b}\right)^2 = \sin^2 \phi \quad \text{----- (6)}$$

1. For $\phi=0$

When phase diff b/n them is zero

$$\left(\frac{x}{a}\right)^2 - \frac{2xy}{ab} + \left(\frac{y}{b}\right)^2 = 0$$

$$\left(\frac{x}{a} - \frac{y}{b}\right)^2 = 0$$

$$y = \frac{b}{a} x$$



Equation of straight line.

The emergent ray is a plane polarized light (PPL) with vibrations in the same plane as that of incident light.

2. For $\phi=\pi$

When Phase diff b/n them is π

$$\left(\frac{x}{a}\right)^2 + \frac{2xy}{ab} + \left(\frac{y}{b}\right)^2 = 0$$

$$\left(\frac{x}{a} + \frac{y}{b}\right)^2 = 0$$

$$y = -\frac{b}{a} x$$



Equation of straight line with

negative slope. The emergent ray is a plane polarized light with direction of vibrations $2\theta = 2 \tan^{-1} \left(\frac{b}{a}\right)$ w.r.t. incident light.

3. For $\phi = (2n-1)\frac{\pi}{2}$ and $a = b$ When phase difference of b/n them
is $\frac{\pi}{2}$ & $\theta = 45^\circ$ w.r.t. O.A.

eq. (7) reduces to

$$x^2 + y^2 = a^2$$



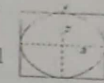
represents an Equation of a circle.

Thus as long as $a = b$, the emergent ray is circularly polarized.

3. For $\phi = (2n-1)\frac{\pi}{2}$ and $a \neq b$ When phase difference of b/n them
is $\frac{\pi}{2}$ & $\theta \neq 45^\circ$ w.r.t. O.A.

eq. (7) reduces to

$$\left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = 1$$

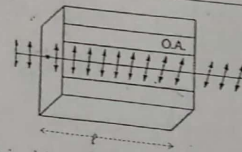


represents an Equation of ellipse.

Thus as long as $a \neq b$, the emergent ray is elliptical polarized.

Optical activity

Incident PPL



When a plane polarized light travels through certain substances like Quartz, turpentine oil, sugar solution etc., its plane of vibration is rotated slowly by some angle θ about the direction of propagation. This property of a crystal is known as Optical Activity & the phenomena as Optical Rotation.

The optical rotation depends upon the thickness ℓ and density of crystal ρ or concentration c (in case of solution), temperature T & wavelength of light used.

Biot's Laws for Rotatory Polarization

The angle of rotation θ of plane of polarization is

1. directly proportional to thickness ℓ of optically active substance
(for any given wavelength of light) $\theta \propto \ell$
2. directly proportional the concentration 'c' of solution
(in case of solution and vapors) $\theta \propto c$
3. If no. of optically active substances are used, the total rotation produced is the sum of rotations produced by different substances independently $\theta_{\text{resultant}} = \sum \theta_i$
4. inversely to the square of wavelength of light passing through it.
5. It is a function of temperature: $\theta = f(T)$ $\theta \propto \frac{1}{\lambda^2}$

Rotatory dispersion

If white PPL is passed through quartz, different colors are rotated through different angles depending on their wavelength

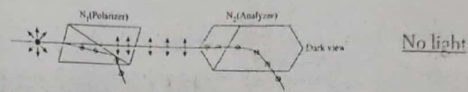
Specific rotation is measure of optical activity of any substance.

$$[\alpha]_T^\lambda = \frac{\theta}{\ell \times c} = \frac{\theta}{\ell \times (m/V)} \quad \text{degree (dm)}^{-1} (\text{g/cc})^{-1}$$

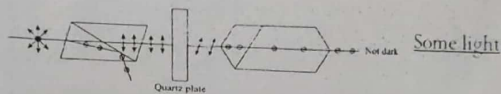
It may be defined as "1 degree of rotation (of plane of polarization of PPL) produced by 1 decimeter length of its solution when concentration is 1 gram/cc (at particular temperature & for given wavelength of light)"

Molecular rotation = Specific rotation $\times$ Molecular weight

Experimental demonstration of optical rotation



If monochromatic light is passed through two Nicol prisms placed in crossed position, no light will emerge out of them.

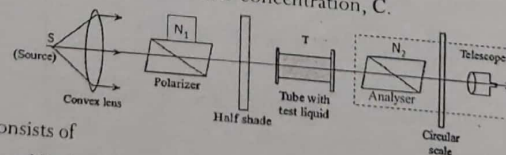


When optically active substance e.g. quartz is placed b/n two crossed Nicols, the field of view is not dark any more, some light emerges out of Nicol N_2 . To restore the original condition of complete darkness, Nicol N_2 has to be turned through some angle. This demonstrates the rotation in the plane of polarization when light passes through the quartz.

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Laurent's half shade polarimeter:

It is an instrument used to measure the angle of rotation, θ produced by certain optically active substances (e.g. sugar solution) & thus can be used to find their specific rotation, S or if specific rotation is known, to find their concentration, C .



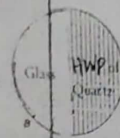
It consists of

- two Nicol prisms, as Polarizer & Analyzer.
- glass tube of certain length, filled with the solution of optically active substance.
- half shade device placed between the polarizer & glass tube.

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Laurent's half plate device

is a circular plate consists of two semi-circular plates of Quartz & glass, cemented together along the diameter YY'.



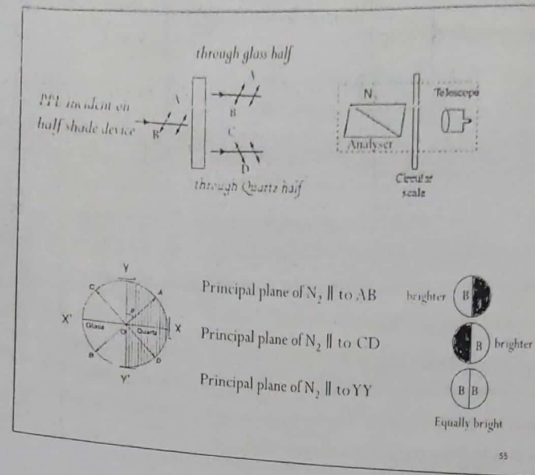
The semi-circular plates on R.H.S. is HWP made of quartz cut parallel to optic axis. It introduces a phase difference of π between o-rays & e-rays on emergence. While semi-circular plate on L.H.S. is a glass plate of certain thickness that it absorbs same amount of light as that absorbed by quartz.

Use: It improves the sensitivity of polarimeter by dividing the field of view into two halves. It provides means for accurately measuring the extent rotation produced by the optically active substance as it helps in relocating the position of equal brightness.

Experiment

The light after passing through polarizer becomes plane polarized with its vibrations in the principal plane of M_1 . Let a plane polarized light is incident normally on half shade device with plane of vibration making an angle θ with optic axis of quartz (along AB).

When light passes through the glass half portion, the plane of vibration of light remains unaffected & hence continues to be along AB. However after passing through quartz HWP the plane of polarization is rotated by 2θ w.r.t. its initial plane. This changes the direction of vibration to along CD at 2θ with respect to the initial vibration along AB.



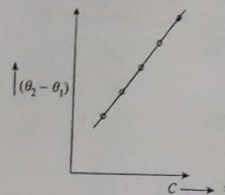
- When Principal plane of N_2 is $\parallel$ to AB, the light emerging from the glass would pass unobstructed through N_2 , while from Quartz half it would be partially obstructed. The field of view from glass half would appear brighter than from Quartz half.
- When Principal plane of N_2 is $\parallel$ to CD, the light emerging from the Quartz half would pass unobstructed through N_2 . The field of view from Quartz half would appear brighter than from glass half.
- When principal plane of N_2 is $\parallel$ to YY', the plane of vibration from both Glass & Quartz portion would be equally inclined w.r.t. the principle plane of N_2 . The field of view over the entire circle will show equal brightness (from both portions).
- When analyzer is slightly rotated from position of equal brightness, marked change in intensity of two halves is observed.

Thus for finding specific rotation of an optically active medium.

- The analyzer reading is noted down (first without solution) for the situation when field of view is equally bright from both portions ("tint of passage" situation) let's say it gives θ_1
- Then by placing an optically active solution in the tube, again analyzer reading for "tint of passage" situation is noted down, say it gives θ_2
- The difference in the angle $(\theta_2 - \theta_1)$ from both cases, gives specific rotation by using equation

$$[S]_T = \frac{(\theta_2 - \theta_1)}{l \times C}$$

degree $(dm)^{-1} (g/cc)^{-1}$



Solve

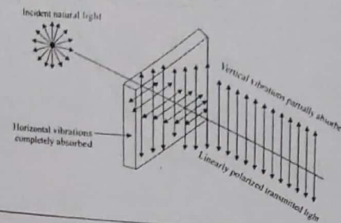
Calculate the minimum thickness of a calcite plate which would convert plane polarized into circularly polarized light. The principal refractive indices are $\mu_o = 1.658$ & $\mu_e = 1.486$ and wavelength of light used $5890 \text{ \AA}$

Solve

Calculate the thickness of HWP for sodium light of $\lambda = 5893 \text{ \AA}$. If $\mu_o = 1.54$ and the ratio of velocity of ordinary and extraordinary waves is 1.007. Is this crystal is negative or positive?

Appendix Polarization by selective absorption

Some double refracting crystals have the property of absorbing strongly one of the two refracting rays, while allowing other to emerge. This selective absorption by the crystal is known as "dichroism". Tourmaline is a naturally occurring crystal which shows this property of selective absorption.



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When unpolarized light is passed through a tourmaline crystal of sufficient thickness, the o-ray is completely absorbed, while the e-ray is almost completely transmitted. So the emergent light is plane polarized.

Polaroid: It is a large sized polarizing film mounted between two glass plates, used to obtain plane polarized light for commercial purposes. The thin film consists of thin sheet of nitro-cellulose packed with ultra microscopic crystals of organic compound with their optic axis all parallel. These crystals are highly dichroic, absorbing one of the doubly refracted beam completely. When unpolarized beam passes through a polaroid, the emergent light is plane polarized. **Uses:** in sunglasses, camera filters, wind screen, widow panes of aeroplane

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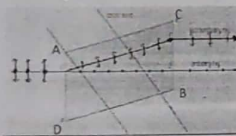
Appendix Huygen's theory of double refraction in uniaxial crystals

Huygen extended his theory of secondary wavelets & explained double refraction phenomena in uniaxial crystals on the basis of wave theory

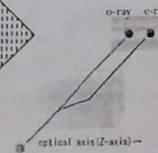
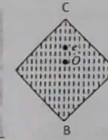
- When light incident on double refracting crystal, every point of it becomes a source of secondary wavelets & sends out two wave fronts, one for o-ray & other for e-ray.
- Wavefront corresponding to o-ray spread out as Spherical as the velocity of o-ray (v_o) remains same in all directions
- while for e-ray, the wavefront is an ellipsoidal of revolution with optic axis as its axis of revolution, since the velocity of e-ray (v_e) is different in different directions in the crystal

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Appendix Experimental visualization of Double Refraction phenomena



Inside the crystal o-ray always lies in the plane of incidence, while e-ray does not.



Mark an ink dot on a piece of paper and place the calcite crystal over this dot. On looking through the opposite face, two images of dots will be observed, corresponding to o-ray & e-ray. On slowly rotating the crystal, it is found that one image rotates around the other image which remains stationary. The stationary image corresponds to o-ray as it obeys law of refraction while the rotating image does not obey the law hence corresponding to e-ray.

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